KCE PATHWAYS · GRADES 4–5 · UNIT 6: VOICES OF CHANGE · WEEK 27 · MATH

The Planetarium Number Hunt

Name: _____ **Date:** _____

PRE-LESSON WARM-UP · DO THIS BEFORE YOUR COMPUTER LESSON

Pre-Lesson Warm-Up: Counting Zeros

Kiona is helping at the Adler Planetarium in Chicago, Illinois, where space numbers have lots of zeros. Warm up before the lesson, and show your work. Some problems may feel new, and that is okay!

1. Review (Week 26): evaluate each expression. a) 5 + 8 × 4; b) 3 × [16 − (2 + 9)]

2. Find each product. a) 10 × 10; b) 10 × 10 × 10; c) 10 × 10 × 10 × 10. What pattern do you see in the zeros?

3. Multiply: 26 × 10 and 26 × 100. Explain what happens to the digits 2 and 6.

4. Which number is 10 times as large as 4.7?

   47 0.47 4.70

5. Kimani says 10 × 10 × 10 × 10 × 10 is 50. Do you agree? Explain your thinking.

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POST-LESSON CHALLENGE · DO THIS AFTER YOUR COMPUTER LESSON

Post-Lesson Challenge: Space Number Check

The exponent counts the tens. Multiplying moves digits to the left, and dividing moves digits to the right. Show your work.

1. Write each in standard form. a) 10^5; b) 7×10^3; c) $4 \times 10^4 + 6 \times 10^2$

2. Multiply or divide. a) 2.36×10^2; b) 0.08×10^3; c) $5{,}400 \div 10^3$; d) $61 \div 10^2$

3. Two steps: a gift shop orders 10^3 moon keychains at $1.25 each and 10^2 planet magnets at $0.90 each. What is the total cost?

4. Find the mistake: Kirra says $3.9 \times 10^2 = 3.900$. In two or three complete sentences, explain her mistake and give the correct product.

★ **Challenge:** 10^7 is how many times as large as 10^3? Then write $1{,}000 \times 10{,}000$ as one power of 10, and explain how you know.

Looking Ahead: Week 28: Multiplying and Dividing Decimals. At a cross-country running meet in Boulder, Colorado, you will multiply and divide decimals to hundredths with times, distances, and pace.